Large Language Model-Based Bidding Behavior Agent and Market Sentiment Agent-Assisted Electricity Price Prediction

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***Abstract*—Day-ahead electricity price prediction is crucial for market participants to make optimal trading decisions. The implementation of the five-minute settlement (5MS) process in the Australian National Electricity Market (NEM) on October 1, 2021, reduced the settlement interval from 30 minutes to 5 minutes. This change has led to more frequent adjustments in pricing, allowing for a more accurate reflection of real-time supply and demand conditions. However, this increased frequency has significantly heightened the complexity of price fluctuations in the wholesale market. Consequently, conventional machine learning and deep learning methods struggle to provide accurate predictions at this higher resolution. Since electricity prices are fundamentally determined by the supply-demand balance and the bidding behaviors of market participants, this work introduces individual participant's bidding behaviors into the prediction model. We fine-tune a pre-trained Large Language Model (LLM) to create bidding behavior agents, which forecasts day-ahead bidding behaviors. Moreover, market sentiment plays a significant role in electricity price volatility, yet it remains challenging to quantify and assess its impact. To address this, we employ a pre-trained LLM to analyze online resources, incorporating market sentiment into the price prediction model. Additionally, to enhance the accuracy of spike predictions, we improve the conditional time series generative adversarial network (CTSGAN) model by utilizing a spike confusion matrix and further strengthen the model by integrating bidding behavior and market sentiment as inputs. Case studies demonstrate that the proposed model significantly improves both electricity price and spike prediction accuracy, offering a robust tool for market participants to navigate the complexities of the modern electricity market.**

***Index Terms*—Large language model (LLM), electricity price prediction, bidding behaviors, market sentiment, spike-enhanced.**

I. Introduction

ELECTRICITY price prediction has attracted significant attention in recent years [1], [2], as accurate price prediction has become crucial for the optimal trading decisions of market participants. These registered participants commonly include 'generators' and 'customers' [3], all of whom rely on price predictions to make critical decisions [4]. 'Generators' typically include power generation companies that rely on coal, gas, wind, solar, and hydro resources to produce electricity. These power generation companies can optimize their market bids based on predicted electricity prices to maximize profits. 'Customers' in electricity markets generally include electricity retailers and large industrial users. Electricity retailers can use price predictions to develop reasonable purchasing strategies and reduce procurement costs. Large industrial users can adjust their production plans based on electricity price predictions to save on electricity expenses. Starting in June 2024, alongside generators and customers, the Australian national energy market (NEM) introduced small resource aggregators (SRA), they can engage in energy trading by aggregating small generation units and storage systems to virtual power plants (VPPs) and energy storage systems (ESSs) [5]. Participants engage in bidding and offering through the spot market, ensuring real-time balance of supply and demand. The settlement process provides a dynamic and responsive mechanism to accommodate fluctuations in electricity supply and demand, thereby ultimately determining the spot price of electricity in the market [6].

In the literature, electricity price prediction most commonly uses time series models [7], include Autoregressive (AR), Moving Average (MA), and Autoregressive Integrated Moving Average (ARIMA) models, which utilize historical price data to forecast future price changes. These models can capture the periodic and trend changes in electricity prices. Additionally, machine learning and deep learning methods are gradually being applied to electricity price prediction. Long Short-Term Memory (LSTM) networks and Gated Recurrent Unit (GRU) models [8], due to their advantages in handling time series data, are increasingly attracting attention from researchers and the industry. These models can learn complex nonlinear relationships in the data, thereby improving prediction accuracy. However, these models have historically been used for price predictions with a resolution of half an hour or more [9].

Since October 1, 2021, the NEM transitioned from a half-hour settlement to a five-minute settlement (5MS). The objectives of this change include: (1) Providing more accurate price signals.

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(2) Promoting market fairness. (3) Improving market responsiveness. (4) Encouraging renewable energy technology innovation and investment [10]. However, the high-frequency settlement has also led to increased competition in the electricity market [11]. Zsuzsanna and Peyman indicated that Australian electricity prices were approximately 4.9% lower due to the implementation of the 5MS program in the NEM [12]. With shorter settlement periods, market participants need to respond more quickly and flexibly to market fluctuations, adding complexity and uncertainty to market operations. This also imposes higher demands on market participants, driving the market towards a more dynamic and competitive environment.

The transition to the 5MS market has also introduced significant challenges for prediction. First, prior to October 1, 2021, although the settlement period was 30 minutes, each 30-minute settlement (30MS) price was derived as the average of six five-minute settlement prices within that period [13]. The shift from 30MS to 5MS means that prices are more prone to sharp spikes, which were often averaged out under the 30MS scheme. Second, in the 5MS market, price fluctuations are more frequent and more sensitive to short-term supply-demand imbalances. The lack of more detailed real-time market information leads to a significant decline in the accuracy of prediction models. This poses greater challenges for traditional models that rely on historical data. Prediction models need to improve their resolution and accuracy to adapt to rapidly changing market conditions and ensure the accuracy of forecasts.

The balance between supply and demand directly determines electricity prices. In the short term, fluctuations in supply and demand in the electricity market have a more immediate and significant impact on prices, driven by factors such as weather changes, unexpected events, and market dynamics. For instance, extreme weather can cause a sudden surge in electricity demand [14], while generation equipment failures or maintenance can lead to sudden supply reductions. These short-term factors not only directly affect the balance of supply and demand but also influence the bidding strategies of market participants. The bidding strategies and subsequent rebidding behavior of these participants often further drive up prices [15]. In a high-frequency settlement environment, where participants must respond more frequently to fluctuations, the complexity and uncertainty of market operations increase. Additionally, their decisions become more susceptible to sudden market news. Therefore, to improve the accuracy of five-minute electricity market prediction models, it is essential to account for both the bidding behaviors of participants and the impact of market news.

Currently, research on bidding behavior in electricity markets mainly focuses on developing optimal bidding strategies from the operator's perspective. These strategies aim to help power generation companies and other market participants maximize their profits while ensuring the stability of power supply and the efficient operation of the market. For example, Tang et al. [16] proposed a novel inverse reinforcement learning (IRL)-based framework to identify the bidding decision objective function for ESS. Christos et al. [17] presented a bi-level optimization framework to determine the optimal bidding strategies for a strategic gas-fired power plant in electricity markets. Many conditions are known to operators, such as their generation curves and optimization models. However, these conditions are confidential to external parties, making it difficult for outsiders to understand the details of their bidding models. Therefore, predicting bidding behavior is extremely challenging. Guo et al. proposed bidding behaviors pattern extraction in electricity markets in [18], and classified three main bidding attitudes (price-preferred, quantity-preferred, and capacity-withholding) using an adaptive clustering method. However, this study does not address the prediction of future bidding behavior. Tang et al. [19] proposed a bidding curve prediction model based on Transformers, which combines feature extraction, encoding, and reconstruction, and achieves accurate prediction results. However, this model is designed for data with a half-hour resolution before the implementation of the 5MS. Since we aim to improve five-minute price predictions using bidding behavior data, this model is not suitable for our application scenario.

Market sentiment is the electricity price volatility driver, Fredj pointed out through a nonlinear high-frequency analysis in [20] that in high-volatility regimes, behavioral factors dominate energy price fluctuations. Market sentiment, which includes participants' reactions to relevant news or events, can lead to more significant and rapid price shifts. For example, extreme weather news is often considered an exogenous variable, helping prediction models understand the impact of weather on renewable energy generation and electricity demand, thereby reflecting in electricity prices [14], [21]. However, it does not help in understanding the role of market sentiment caused by extreme weather on electricity prices. Furthermore, news is often presented as long texts, which are difficult to directly use as exogenous inputs. These texts contain a wealth of details and background information that prediction models struggle to effectively utilize without proper processing. Most studies resort to counting keyword frequencies [20] or treating events as binary variables [22], but this approach fails to grasp the nuanced meanings and complex features underlying the text. As a result, traditional methods cannot effectively handle complex news texts nor capture the impact of market sentiment on electricity prices.

In recent years, Large Language Models (LLMs), such as the GPT [23] and LLaMa [24], have emerged as a cutting-edge technology in artificial intelligence, demonstrating remarkable capabilities across various tasks. The substantial scale of LLMs grants them extensive computational power. Additionally, pre-training on large-scale datasets endows these models with broad general knowledge and refined reasoning abilities, allowing them to capture complex patterns and dependencies, making them powerful tools for building intelligent systems with a degree of understanding. The architecture of LLMs is increasingly being applied to tasks beyond language, such as time series processing [25], [26]. For example, Tian et al. [27] demonstrated that frozen LLMs can effectively perform in time series tasks by capitalizing on the adaptability of self-attention mechanisms. Furthermore, advancements in reasoning with LLMs, such as chain of thought [28] prompting, enable step-by-step emulation of human reasoning processes. The tree of thoughts [29] approach further enhances this by introducing a trial-and-error mechanism with multi-round dialogue and memory feedback modules to improve decision-making. The combination of scale, extensive data pre-training, and emergent reasoning abilities positions LLMs as powerful tools for complex predictive tasks across domains [30], making it possible for them to learn and predict bidding behavior for electricity market participants and to interpret market sentiment [31] and analyze its impact on electricity prices.

In our previous work [32], we proposed the conditional time series generative adversarial network (CTSGAN) prediction model for wholesale market electricity price prediction. While it performed well overall, its ability to predict spikes remained insufficient. Accurate spike prediction is crucial for new market participants in the energy sector, including VPPs and ESSs. These participants rely on precise spike predictions to maximize profits through rapid charging and discharging. Improving the accuracy of spike predictions remains a significant challenge.

To enhance the accuracy of electricity price prediction models, several key challenges must be addressed: (1) Predicting future bids is crucial since the bidding behavior of market participants directly influences prices. (2) Market sentiment plays a significant role in price volatility, yet it remains difficult to analyze effectively. (3) The precision of spike predictions still requires improvement. The main contributions of this work are as follows:

1) The supply and demand dynamics in the electricity market drive price fluctuations. This work proposes a fine-tuning LLM called the bidding behavior agent, which simulates the five-minute bidding behavior of both scheduled and semi-scheduled market participants.
2) Market sentiment significantly impacts electricity price volatility. We introduce a market sentiment agent, which can analyze various market news and accurately estimate its impact on hourly electricity prices for the next day.
3) An improved spike-enhanced CTSGAN model, incorporating a confusion matrix, is introduced to refine spike prediction accuracy. Based on this, we propose a comprehensive framework that integrates the bidding behavior agent and market sentiment agent, resulting in more accurate electricity price predictions.

The rest of this paper is organized as follows: Section II investigates fine-tuning LLM-based bidding behavior agent. Section III introduces pre-trained LLM-based market sentiment agent. Section IV presents the two agent-assisted spike-enhanced CTSGAN prediction framework. Section V conducts the case studies, followed by the conclusion in Section VI.

II. Bidding Behavior Agent for Price Band Prediction

A. Settlement Mechanism in Wholesale Market

The conventional electricity prediction models typically ignore the bidding process and directly treat wholesale electricity prices as a time series. However, in reality, the determination of electricity prices is complex and depends on both the demand and supply of electricity. Therefore, it is necessary to consider the behavior of the participants. The settlement process includes the bidding and clearing stages [30].

In the bidding stage, the NEM operates a real-time spot electricity market, allowing participants to submit bids. Scheduled and semi-scheduled participants must submit their price bands (PBs), a total of 10, before 4:00 am each trading day, and these prices must be monotonically increasing. Once the trading day begins, these 10 price bands remain fixed and cannot be adjusted. In the real-time trading stage, participants can frequently update the band availabilities (BAs) of each PB throughout the day, with each price corresponding to one of the 10 BAs. However, participants must submit their final BA values at least 5 minutes before the clearing stage. For example, for an 8:10 am electricity trade, participants must adjust and submit their final BA before 8:05 am. During the market clearing stage, AEMO matches buyers and sellers to meet electricity demand with the least cost. Therefore, the lowest-priced offers and the highest-priced bids are prioritized. The final clearing price is determined by the highest offer among all successful transactions, ensuring that only the most competitive offers are executed.

B. Day-Ahead Five-Minute Band Availability (BA) Prediction

In Australian NEM, participants in spot electricity trading can primarily be divided into two categories: generators and customers, corresponding to sellers and buyers of electricity, respectively. With the development of the electricity market, SRAs have also begun participating in spot trading. These SRAs act as customers when charging (buying electricity) and as generators when discharging (selling electricity).

Since the implementation of 5MS process in October 2021, the volume of each settlement has become smaller, making participants' bidding behavior more likely to influence price fluctuations, thereby increasing the randomness of electricity prices [12]. Therefore, it is necessary to analyze participants' bidding behavior to achieve more accurate electricity price forecasting.

Participants typically have professional analysis teams that analyze factors such as generation costs, market demand, and electricity prices to submit optimal offers and bids and achieve the best possible profits. Their bidding behavior are not merely the result of simple mathematical models but are deeply influenced by the strategies of the traders. Since the underlying logic behind these strategies is unknown to us, conventional models are often inadequate for fitting this decision-making process.

Based on the statistics of participants' historical PBs and BAs, we found that participants rarely modify their PBs. For example, Bayswater Power Station, which is a bituminous coal-powered thermal power station with a capacity of 2640 MW, only changed its PBs 10 times from October 2021 to June 2024. At the same period, the Tumut 3 Power Station (1800 MW), only changed its PBs 3 times. Due to the minimal data, it is challenging to effectively uncover the underlying patterns of PBs. Therefore, for predicting the PBs, we use a basic naive method [33], assuming that the PBs on the prediction day be the same as the previous day. The bidding behavior predictions discussed in this work focus solely on BAs. Additionally, AEMO publishes the details of the previous trading day's bidding data at 4:00 AM each day. Therefore, although the bids are settled every

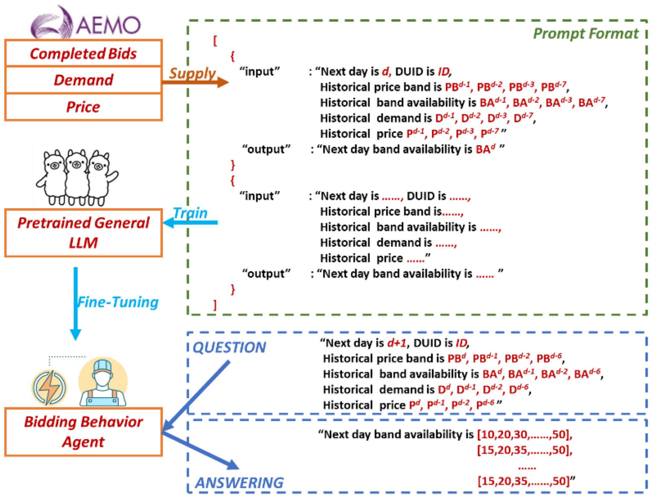


Fig. 1. Framework of LLM-based behavior agent for future bid prediction.

5 minutes, the detailed bidding information is not available in real-time. As a result, our work focuses on the prediction of the day-ahead BAs for each participant.

C. LLM-Based Bidding Behavior-Agent

Language models are trained on a collection of sequences $\mathcal{U} = \{U_1, U_2, \ldots, U_n\}_{n=1}^{N}$, where $U_n = (u_1, u_2, \ldots, u_j, \ldots, u_{n_i})$ and each u_j is a token representing word or parts of words. These tokens are part of a vocabulary $\mathcal{V}$ and are generated by a pre-defined tokenizer. LLM typically encode an autoregressive distribution, where the probability of each token depends solely on the preceding tokens in the sequence:

$$p_\theta\ (U_n) = \prod_{j=1}^{n_i} p_\theta(u_j | u_{0:j-1}) \tag{1}$$

where θ represents the model parameters, which are learned by maximizing the likelihood of the entire dataset $p_\theta(\mathcal{U}) = \prod_{n=1}^{N} p_\theta(U_n)$. The sequence $u_{0:j-1}$ denotes the tokens from 0 to the $j-1$ token. We employ a trained language model using a sampling method, starting with a prompt $u_{0:k}$ and generating tokens sequentially using $p_\theta(u_j|u_{0:j-1})$.

LLMs are traditionally used for predicting word tokens, but with billions of parameters, they excel at representing complex probability distributions over sequences [34], making them well-suited for uncovering the underlying factors behind each electricity market participant's bidding and predicting future bidding behavior. Fig. 1 illustrates the framework for fine-tuning a pre-trained general LLM into a bidding behavior agent and using the agent for BA prediction. Firstly, we collect publicly available datasets from AEMO, including historical bidding data, historical electricity price data, and historical electricity demand data, along with each participant's Dispatchable Unit Identifier (DUID) number. The historical bidding data includes historical PB data, and historical BA data. The format of the prompt is given in Fig. 1.

The input includes DUID, prediction date d, historical PBs, where each PB is a 1×10 array, historical BAs, where each BA is a 288×10 array, historical demand D, and historical electricity spot price P, both of which are 288×1 arrays. The above PB, BA, D, and P respectively include data for dates d-1, d-2, d-3, and d-7. The output includes the BA for d, which is a 288×10 array. One input and output as described above constitute one sample. We use all historical data from October 8, 2021, to April 30, 2024, as the fine-tuning training set, with a total of 934 samples.

We perform fine-tuning on pre-trained LLM using the training set through a parameter-efficient method called Low-Rank Adaptation (LoRA). The core concept of LoRA is to conduct low-rank adaptation on the self-attention module and weight matrix of feed-forward neural network layers in the Transformer. Therefore, the original weight matrix remains unchanged while only the newly added low-rank matrix is updated, which significantly reduces the number of trainable parameters and computation required. LoRA typically applies to the query W^Q and key W^K matrices of the self-attention layer, as well as the weight matrix of the feed-forward neural network layers W^f. Assume the original weight matrix is $W \in \mathbb{R}^{l \times n}$, where l is the output dimensionality and n is input dimensionality. LoRA introduces two low-rank matrices $A \in \mathbb{R}^{l \times m}$ and $B \in \mathbb{R}^{m \times n}$, where m represents the rank of a low-rank matrix, which is much smaller than l and n. The original weight matrix W is updated to $W + AB$, where the calculation of AB represents a low-rank approximation. By adding two small matrices, A and B, as trainable parameters, LoRA greatly reduces the number of parameters required for fine-tuning, making it more efficient when dealing with large models. LoRA only adjusts a subset of the model's weights and adds low-rank matrices to capture features specific to the new task, rather than updating all model parameters. This selective updating allows the model to retain most of its pre-trained knowledge while avoiding overfitting to the characteristics of the fine-tuning data for a specific task. Furthermore, by reducing the number of parameters that are updated, LoRA enables the model to focus on the core features of the new task without over-learning minor patterns in the training data, thereby reducing the risk of overfitting during fine-tuning.

By leveraging LoRA, we can efficiently fine-tune the large language model, transforming the general LLM into a professional bidding expert for the electricity market, thereby predicting future BA. Additionally, this bidding behavior agent can learn the different bidding styles and strategies of each participant through their DUID, ensuring a more accurate simulation of the bidding scenarios in the electricity market.

III. Market Sentiment Agent for Estimating Impact on Electricity Prices

In the previous section, we utilize an LLM-based bidding behavior agent to predict the future actions of market participants, allowing us to understand the impact of bidding behavior on electricity prices from the perspective of individual participants. In this section, we will take a more holistic approach to understanding the changes in market sentiment and their influence on electricity prices.

A. Market Sentiment-Related Factors

Market sentiment-related factors [35], [36] refer to elements that can influence the overall psychological expectations and behavior within the market. These factors are equally important in the electricity spot market, including: (1) Weather Conditions: Extreme weather (such as heatwaves, cold snaps, storms) affect electricity demand and supply, influencing market sentiment. This is especially significant during peak seasons like summer and winter. (2) Market News: News reports related to the electricity market can influence participants' psychological expectations. This includes news about power plant shutdowns, infrastructure failures, market reforms, etc. (3) Policy and Regulatory: New policies or regulations, particularly those related to energy, environment, and market structure, have a significant impact on market sentiment. For example, adjustments to renewable energy subsidies or changes in carbon emission policies. (4) Economic Indicators: GDP growth rate, inflation rate, unemployment rate, can affect market confidence. While these indicators are often associated with long-term trends, they can also have immediate effects on short-term electricity prices. For example, a sudden report of unexpected inflation or a sharp rise in unemployment can lead to a rapid shift in market sentiment, causing participants to reassess their expectations for near-term electricity demand and adjust their bids and trading strategies accordingly.

The impact of sentiment-related factors on prices is complex, involving interactions between various elements, such as the mutual influence of economic indicators and policies. Market sentiment is also shaped by electricity prices, with price fluctuations altering overall market expectations and behavior. The effects of these factors on electricity prices are often non-linear; for instance, extreme weather can have a much stronger impact during peak usage periods. The complexity of behavioral finance means that the same factors can trigger different market sentiments. Additionally, sentiment-related factors influence electricity prices over varying time scales, with timely news and weather forecasts often having limited long-term effects. These complexities make it challenging to directly quantify market sentiment.

B. LLM-Based Market Sentiment Agent

Market sentiment is the general attitude or outlook of electricity market participants, reflecting collective emotions and opinions about future market conditions [31]. It is often influenced by qualitative factors such as news events, economic indicators, and shifts in public opinion. Market sentiment can drive trends and price movements, making it a valuable indicator in forecasting models, especially when traditional numerical data may not fully capture the nuances of current events or societal changes. We use S to represent the market sentiment factor, capturing the impact of weather conditions WC, electricity market news MN, policy and regulatory PR, and economic indicators EI on future

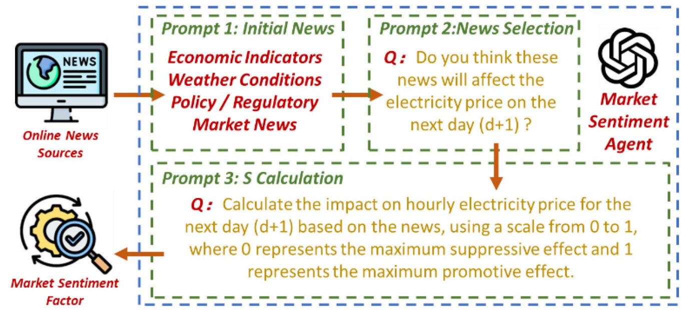


Fig. 2. Framework of LLM-based market sentiment agent for next day electricity market.

electricity market sentiment:

$$S = f\,(WC, MN, PR, EI) \tag{2}$$

where f is a nonlinear function that combines the related factors in a complex way. Therefore, LLM is introduced to calculate the market sentiment factor S for future electricity market sentiment based on economic, weather, policy and regulatory, and electricity market news.

The framework illustrated in Fig. 2 describes the process of utilizing a LLM to calculate hourly market sentiment factor that predicts the impact of news on electricity prices for the next day. The process begins with the collection of diverse news articles from online sources ABC News and news.com.au, including economic indicators, weather conditions, policy and regulatory, and electricity market news. These articles are initially processed by the LLM, which evaluates their content through a prompt (prompt 2) asking whether the news is expected to affect the electricity price on the following day *d+1*. Relevant news is then selected for further analysis. In the final step (prompt 3) the LLM calculates the impact on the hourly electricity price for the next day using a scale from 0 to 1, where 0 indicates the maximum suppressive effect and 1 indicates the maximum promotive effect. The resulting market sentiment factor S is a 24×1 array, which quantifies the expected influence of the news on future hourly electricity prices for the next day, thereby providing valuable insights for electricity price forecasting.

IV. Spike-Enhanced CTSGAN Prediction Model

A. CTSGAN Model

The basic GAN model is mainly composed of two neural networks, the generator $G(\cdot)$ and the discriminator $D(\cdot)$. The responsibility of the generator is to generate a massive number of realistic scenarios with different random noise inputs z under a well-defined noise distribution $Z \sim \mathbb{P}_z$, e.g., Gaussian distribution, until the discriminator cannot distinguish them from the real scenarios x. The discriminator's responsibility, essentially a classifier, is to distinguish the generated and the real historical scenarios. Denote the generator neural network and discriminator neural network as $G(z;\theta^{(G)})$ and $D(z;\theta^{(D)})$, and the weights of neural networks in the generator and discriminator as $\theta^{(G)}$ and $\theta^{(D)}$, respectively. The ideal output $G(z;\theta^{(G)})$ should follow the distribution of real historical data $X \sim \mathbb{P}_x$ after being

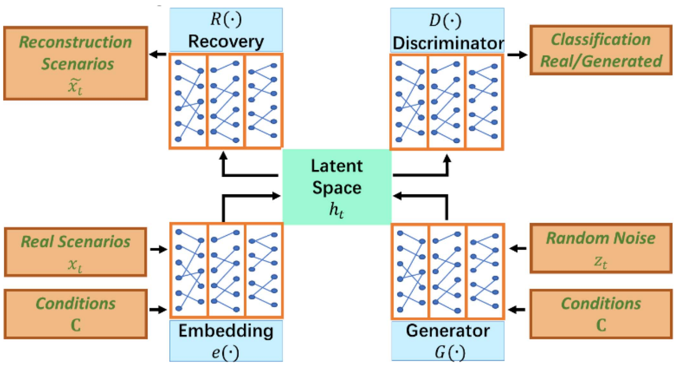


Fig. 3. Structure of the CTSGAN model.

well trained. The discriminator is trained to identify $\mathbb{P}_x$ from $\mathbb{P}_z$, and thus to maximize the difference between $D(x;\theta^{(D)})$ and $D(G(z;\theta^{(G)});\theta^{(D)})$. According to the objectives of the generator and the discriminator, loss functions L_G and L_D are formulated to train and optimize the weights of the neural networks of the generator and the discriminator, respectively. A small L_G indicates that the generator can synthesize realistic scenarios, which as if comes from the real historical scenarios. A small L_D reflects that the discriminator cannot identify whether the generated scenarios are from the historical scenarios. Define the L_G and L_D as:

$$L_G = -\mathbb{E}_Z\left[log\left(1 - D\left(G\left(z;\theta^{(G)}\right);\theta^{(D)}\right)\right)\right] \tag{3}$$

$$\begin{aligned} L_D = & -\mathbb{E}_X\left[log\left(D\left(x;\theta^{(D)}\right)\right)\right] \\ & -\mathbb{E}_Z\left[log\left(1 - D\left(G\left(z;\theta^{(G)}\right);\theta^{(D)}\right)\right)\right] \end{aligned} \tag{4}$$

where $\mathbb{E}_X$ and $\mathbb{E}_Z$ denote the expected value over all real scenarios and the expected value over all random noise input to the generator, respectively.

Based on the basic GAN, we proposed the CTSGAN prediction model in our previous work, which achieved good results in both point prediction [32] and interval prediction [21] of electricity prices. The structure of the CTSGAN model can be found in Fig. 3.

Different from conventional GAN, the proposed CTSGAN model contains two additional sections: an embedding section $e(\cdot)$ and a recovery section $R(\cdot)$. It can map between feature and latent space and help GAN learn the underlying temporal dynamics of the training scenarios via low-dimensional representations.

$$h_t = e(h_{t-1}, x_t|\mathbf{C};\theta^{(e)}) \tag{5}$$

$$\widetilde{x}_t = R\left(e\left(h_{t-1}, x_t|\mathbf{C};\theta^{(e)}\right);\theta^{(R)}\right) \tag{6}$$

where x_t and h_{t-1} are the inputs of the embedding neural network at time t, h_t denotes output of the embedding neural network at time t; $\theta^{(e)}$ and $\theta^{(R)}$ are the weights of the embedding and recovery neural networks, respectively. $\mathbf{C}$ represents the price-related conditions. As $e(\cdot)$ and $R(\cdot)$ should enable accurate reconstruction of the original data from latent space, the first objective is the reconstruction loss expressed as:

$$L_R = \mathbb{E}_X\left[\sum_t\left(\left\|R\left(e\left(h_{t-1}, x_t|\mathbf{C};\theta^{(e)}\right);\theta^{(R)}\right) - x_t|\mathbf{C}\right\|_2\right)\right] \tag{7}$$

where $\|\cdot\|_2$ denotes the Euclidean norm, and $\theta^{(e)}$ and $\theta^{(R)}$ denote the weights of the two neural networks in embedding and recovery functions, respectively. In open-loop mode, the second objective is the unsupervised loss expressed as (8), which we use a linear function [37] to replace the logarithmic function to accelerate and stabilize the training process.

$$\begin{aligned} L_U = & \mathbb{E}_X\left[\sum_t\left(D\left(e\left(h_{t-1}, x_t|\mathbf{C};\theta^{(e)}\right);\theta^{(D)}\right)\right)\right] \\ & -\mathbb{E}_Z\left[\sum_t\left(D\left(G\left(h_{t-1}, z_t|\mathbf{C};\theta^{(G)}\right);\theta^{(D)}\right)\right)\right] \end{aligned} \tag{8}$$

In the closed-loop mode, the generator receives the embedding network output to generate the next latent vector, and thus, the third objective is the supervised loss expressed as (9):

$$\begin{aligned} L_S = \mathbb{E}_{X,Z}\Bigg(\sum_t & \left\|e\left(h_{t-1}, x_t|\mathbf{C};\theta^{(e)}\right)\right. \\ & \left.- G\left(h_{t-1}, z_t|\mathbf{C};\theta^{(G)}\right)\right\|_2\Bigg) \end{aligned} \tag{9}$$

B. Spike Enhancing With Confusion Matrix-Based Loss

Prior to October 2021, the 30MS often resulted in short-duration spikes being averaged into non-spike data, which led to a low proportion of spikes. Traditional prediction methods tended to ignore spikes and focus solely on predicting regular prices. However, with the implementation of 5MS, the proportion of spikes has increased significantly. This means that when designing prediction methods, we need to account for spikes and even emphasize their weight in prediction error calculations.

The CTSGAN prediction model demonstrates accurate performance in point prediction of electricity prices. However, accurately predicting spikes remains a challenging task. In our previous work [38], we set a threshold for spikes using a confusion matrix. However, the method did not improve spike prediction accuracy. Inspired by these limitations, this work directly incorporates confusion matrix metrics related to spike prediction into the supervised loss function, aiming to enhance prediction accuracy by explicitly addressing spike-related errors. This addition to the supervised loss function ensures that the model pays specific attention to correctly identifying spikes while reducing the occurrence of false spike predictions. Therefore, the second objective is modified as spike-enhanced supervised loss L_S^{SE}, can be expressed as:

$$L_S^{SE} = L_S + \lambda_{CM} \cdot L_{CM} \tag{10}$$

where L_{CM} represents the confusion matrix loss for spikes, and λ_{CM} is a weight parameter that balances the supervised loss L_S and the confusion matrix loss L_{CM}. The confusion matrix loss

L_{CM} can be defined as:

$$L_{CM} = FPR + FNR \tag{11}$$

$$FPR = {}^{FP}/_{(FP+TN)} \tag{12}$$

$$FNR = {}^{FN}/_{(FN+TP)} \tag{13}$$

where FPR and FNR represent false positive rate and false negative rate [39], defined as (12) and (13). TP denotes True Positive, meaning correctly predicted spikes; FP denotes False Positive, meaning incorrectly predicted spikes; TN denotes True Negative, meaning correctly predicted non-spikes; FN denotes False Negative, meaning incorrectly predicted non-spikes.

In this work, we set the threshold for spikes as the cap price of the Australian electricity market, which is 600 AUD/MWh. This upper limit reflects price peaks driven by extreme demand or supply shortages, making it a reasonable standard for identifying spikes. Additionally, the cap price, as a critical point within the market mechanism, helps identify abnormal price fluctuations over short periods, enabling the forecasting model to more accurately capture spike events that significantly impact market volatility.

By incorporating the confusion matrix-based loss into the CTSGAN model, we aim to enhance the model's ability to predict spikes, thereby improving its overall predictive performance for electricity prices. This approach ensures that the model remains robust in generating accurate predictions while specifically targeting the challenges associated with spike prediction.

C. LLM-Agent-Assisted Electricity Price Prediction

By fine-tuning the LLM-based bidding behavior agent in Section II, the bidding strategies for different DUID participants can be learned, thereby predicting the five- minute BAs of each participant for next day, denoted as $BA_{288\times10}^{d+1,\ DUID}$. Additionally, in Section III, filtering and selecting historical news are conducted using a pre-trained LLM-based market sentiment agent to obtain the impact on hourly electricity prices for the next day, referred as the future market sentiment factor $S_{24\times1}^{d+1}$. Fig. 4 presents the electricity price prediction framework with future $BA_{288\times10}^{d+1,\ DUID}$ and $S_{24\times1}^{d+1}$ from the two LLM-based agents.

In addition to $BA_{288\times10}^{d+1,\ DUID}$ and $S_{24\times1}^{d+1}$, the conditions input for CTSGAN also includes historical electricity prices and historical demand from *d, d-1, d-2, d-6*, and *d-13*, all with a five-minute resolution. The conditions also include the forecasted electricity prices and forecasted demand for *d+1* provided by AEMO, with a resolution of 30 minutes. Since these inputs have inconsistent resolutions, it is necessary to upscale all resolutions to 5 minutes using linear interpolation. Finally, we denote the 5-minute resolution data as: $BA_{288\times10}^{d+1,\ id}$ for the participant's band availability with DUID of *id*; $S_{288\times1}^{d+1}$ for future market sentiment; $P_{288\times1}^{d}$, $P_{288\times1}^{d-1}$, $P_{288\times1}^{d-2}$, $P_{288\times1}^{d-6}$, $P_{288\times1}^{d-13}$ for historical spot electricity prices; $D_{288\times1}^{d}$, $D_{288\times1}^{d-1}$, $D_{288\times1}^{d-2}$, $D_{288\times1}^{d-6}$, $D_{288\times1}^{d-13}$ for historical demand; $P_{288\times1}^{d+1,AEMO}$ and $P_{288\times1}^{d}$ for predicted price and demand by AEMO for $d+1$.

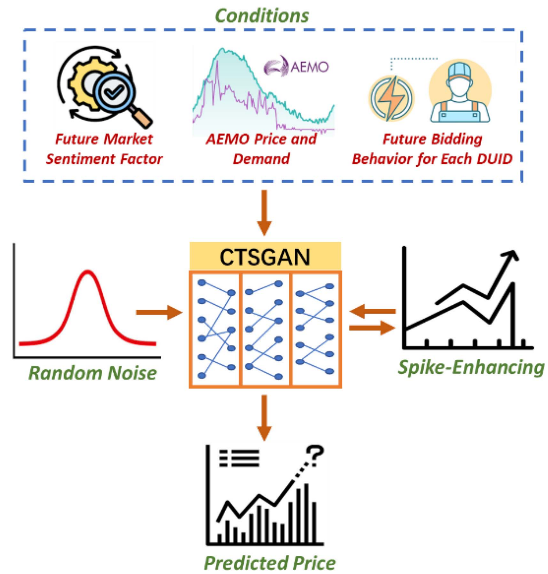


Fig. 4. Electricity price prediction framework.

Additionally, we select 600 AUD/MWh as the threshold for spikes, as the current Australian administered price cap [40] is 600 AUD/MWh. The conditions and random noise are used as inputs for training CTSGAN. Moreover, the model is trained using a spike confusion matrix to enhance the training specifically for the spike parts. Finally, we obtain a pre-trained prediction model, which can be directly for day-ahead 5 min electricity price prediction.

V. Case Study and Result Analysis

In this section, we use the spike-enhanced CTSGAN model proposed in this work to predict day-ahead electricity prices with a 5-minute resolution, incorporating bidding behavior and market sentiment. We introduce the implementation details firstly. Then, we validate the effectiveness of the bidding behavior predictions with fine-tuning LLM-based bidding behavior agent. Subsequently, we assess the prediction performance of the spike-enhanced CTSGAN for both spike and non-spike electricity prices. Moreover, ablation experiments are conducted to compare the effects of bidding behavior and market sentiment on prediction performance. Finally, we compare our proposed model with conventional models to further demonstrate its effectiveness.

A. Implementation Details

The training dataset of 5 min electricity spot prices and demands in New South Wales, Australia, from October 1, 2021 is downloaded from Australian Energy Market Operator (AEMO). The threshold for spike price is set at 600 AUD/MWh. The core neural networks and hyperparameters for the proposed CTSGAN are consistent with those used in [21], [32]. These parameters have been thoroughly optimized through comparative experiments conducted in prior work to identify the most effective configurations. All predictions with CTSGAN are implemented in Python 3.8 with the packages of Pytorch on

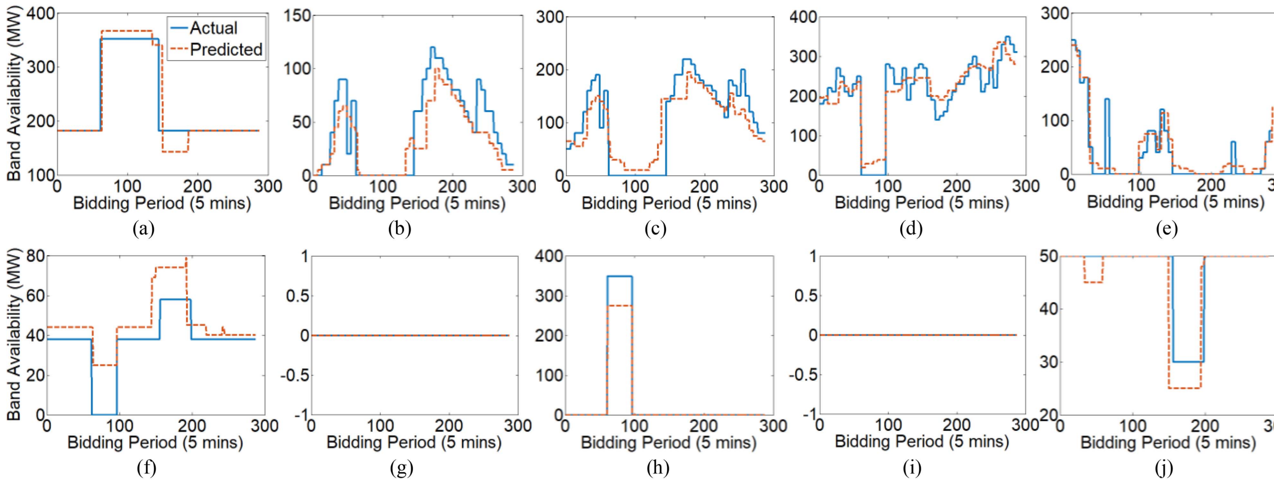


Fig. 5. Day-ahead BA predictions with bidding behavior agent, (a)–(j) representing PB1-PB10.

VSCode and run on a PC with an AMD RYZEN 9 3950X CPU and an NVIDIA RTX 3080 GPU.

AEMO releases the latest electricity price and demand data in real time. In addition, AEMO updates bid datasets daily, which include historical PB and BA data. According to the description in Section II-C, the fine-tuning dataset starts from October 8, 2021, with a total of 934 samples. An example of input for fine-tuning the LLM-based bidding behavior agent is shown in Appendix A. The bidding behavior agent utilizes the LLaMa3 8B pre-trained model, which comprises 8 billion parameters. The fine-tuning process is trained using the LLaMa-Factory tool. The fine-tuning method selected is LoRA, with the stage set to supervised fine-tuning. The learning rate is set to 5e-5, with 3.0 epochs, a maximum gradient norm of 1.0, a batch size of 2, gradient accumulation of 8, and a cutoff length of 1024. All other settings are kept at their defaults. The choice of the above parameters allows the model to learn gradually, helping to maintain stability and avoid overly large updates that could lead to overfitting. Additionally, during training, we rely on the convergence curve provided by LLaMA-Factory to constantly monitor the loss. If there is no further improvement, we stop the training, effectively preventing the model from learning noise in the data, which could lead to overfitting. The fine-tuning bidding behavior agent is conducted with NVIDIA A5000 24G GPU.

The market sentiment agent takes as input the online news sources from the past 20 days, including ABC News and news.com.au. News articles are initially screened by comparing headlines with a keyword library (the keyword word cloud is provided in Appendix B). For the market sentiment analysis, we leverage the capabilities of ChatGPT-4o, which serves as a pre-trained market sentiment agent due to its advanced natural language processing capabilities and extensive pre-training on diverse datasets.

B. Verification of BA Prediction With Bidding Behavior Agent

This work takes the electricity market participant with DUID ER01 as an example to conduct the day-ahead BA prediction for May 20, 2024. ER01 is a 720 MW Toshiba steam-driven turbo-alternator at Eraring Power Station, a coal-fired power station owned by Origin Energy. Fig. 5 presents the BA prediction results based on the LLM-based bidding behavior agent, where (a) to (j) represent the 10 BAs corresponding to the 10 PBs. The horizontal axis represents the 288 5-minute bidding periods for each trading day, and the vertical axis represents BAs. The blue solid line indicates the actual BAs, while the brown dashed line represents the predicted BAs.

It can be observed that the LLM-based bidding behavior agent's predictions are generally close to the actual values, capturing the overall trends in BAs. Even in subfigures (b), (c), and (d), the predictions accurately capture the significant changes, especially around the bidding periods 50 and 170, which correspond to the commonly peak spot prices in the morning and evening, respectively. For subfigure (a), ER01 provides approximately 350 MW of BA around the 100th bidding period at a lower PB. Our behavior agent is also able to learn this typical strategy, which involves bidding at lower prices during the midday to afternoon period when solar output is high and demand is insufficient. Additionally, as seen in subfigures (g) and (i), ER01 did not submit any BA for PB7 and PB9. The proposed agent predicts this strategy with 100% accuracy.

We introduced conventional time series prediction models, GRU and LSTM, configured with a activation function of tanh, a recurrent activation function of sigmoid, a hidden layer size of 300, and 2 layers, to predict BA1-BA10 and compared their RMSE errors, as shown in Fig. 6. It can be observed that relying solely on historical time series prediction methods is quite inaccurate for BA prediction. For example, for BA1, the RMSE of the bidding behavior agent constructed with LLaMa3 8B is only 29.92, while the RMSE of GRU is as high as 193.51, and the RMSE of LSTM is 86.27. Additionally, from BA7 and BA9, it can be seen that the LLaMa3 8B-based agent can accurately predict that the BA for PB7 and PB9 is 0, whereas the errors obtained from relying solely on time series prediction are very significant. This further demonstrates that the LLaMa3

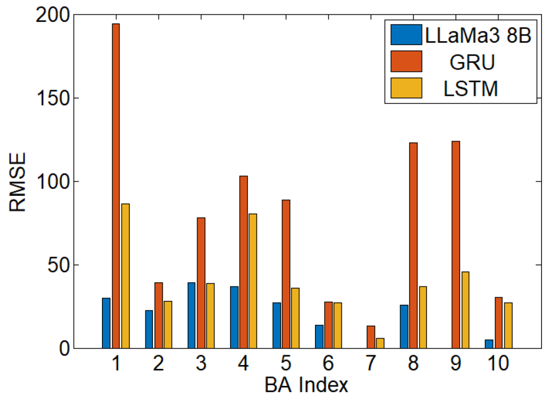


Fig. 6. RMSE for day-ahead BA predictions with different prediction models.

TABLE I
COMPARISON OF FINE-TUNING/TRAINING AND INFERENCE/TEST TIMES FOR DIFFERENT BA PREDICTION MODELS

	Fine-tuning/ Training (Second)	Inference/Test (Second)
LLaMa 3 8B	179224.62	6.35
GRU	1124.61	2.23
LSTM	1280.95	2.47

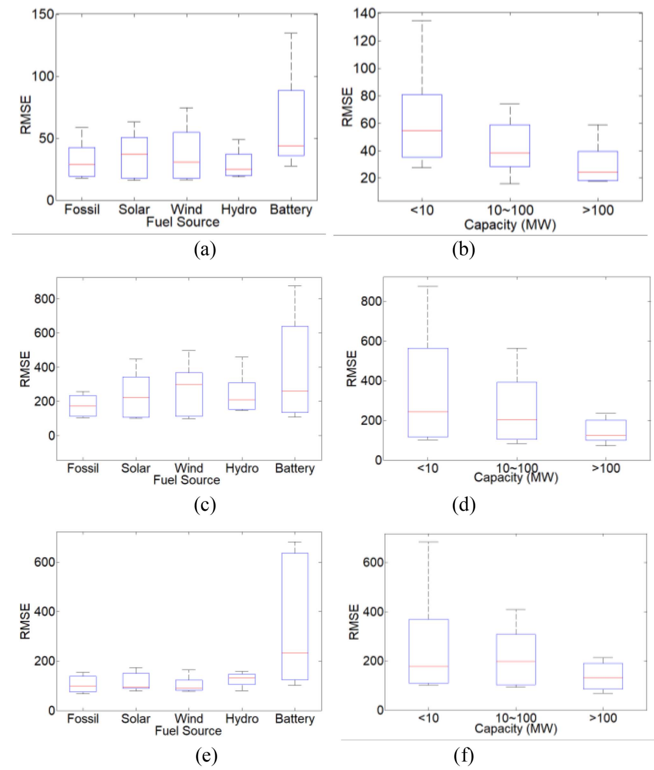


Fig. 7. Box plots of RMSE for day-ahead BA predictions in terms of (a), (c), (e) fuel source and (b), (d), (f) capacity size with (a), (b) LLaMa 3 8B, (c), (d) GRU and (e), (f) LSTM.

8B-based agent is not just performing time series prediction but is also conducting strategy prediction for bidding behavior. Table I presents the fine-tuning and inference times for LLaMa 3 8B, compared with the training and testing times of the LSTM and GRU models. It can be observed that the fine-tuning time for LLaMa 3 8B is 179224.62 seconds (approximately 49 hours), whereas the training times for LSTM and GRU are under 20 minutes, showing a significant disparity. The primary reason for this difference is that LLaMa 3 8B contains 8 billion parameters, which greatly increases the fine-tuning time cost. However, when comparing inference and test times, all models require under 10 seconds, with no significant order-of-magnitude difference. Additionally, the bidding behavior agent does not require daily fine-tuning; typically, the fine-tuning frequency can be adjusted to once per week or even once per month.

Fig. 7 shows box plots of RMSE for day-ahead BA predictions categorized by fuel source and capacity size with LLaMa3 8B, GRU and LSTM. Fig. 7(a) and (b) present the BA prediction results using LLaMa3 8B. It can be observed from (a) that the RMSE varies across different fuel sources, with fossil fuels and wind showing relatively lower errors compared to batteries, which exhibit the highest variability and median RMSE. This suggests that BA predictions are more accurate for certain fuel types, likely due to the more predictable nature of their output. Additionally, as a new participant, most batteries entered the electricity market after 2022. The lack of sufficient operational experience has led to inconsistent strategies over time, significantly impacting predictions. Furthermore, the lack of adequate training data for the prediction models is another major factor contributing to the high error rates. One can find that the RMSE also differs based on capacity size from (b). Predictions for capacities less than 10 MW have higher variability while capacities greater than 100 MW exhibit small RMSE values. This indicates that larger generators tend to have more consistent bidding strategies, making it easier for the bidding behavior agent to learn.

Compared to (a), the BA prediction results provided by GRU in (c) and by LSTM in (d) show a significant increase in error. The average RMSE of fossil fuel market participants in (a) is approximately 34, while in (c) for the same participants, the RMSE reaches 173, and in (e) it is 102. Additionally, for battery participants with insufficient data, the maximum RMSE for GRU is as high as 876, and for LSTM, it is 681, both far exceeding the 138 of LLaMa 3 8B. This indicates that LLaMa 3 8B performs well even for new participants with limited data and unstable strategies. Furthermore, comparing (d) and (f) with (b), it can be observed that LLaMa 3 8B consistently demonstrates significant advantages over GRU and LSTM for power market participants across different capacity types.

C. *Verification of Normal Electricity Price Prediction With Two LLM Agent*

In this section, we introduce ablation experiments to analyze the impact of the bidding behavior agent and the market sentiment agent on day-ahead normal price prediction (<600AUD/MWh). It should be noted that the prediction model

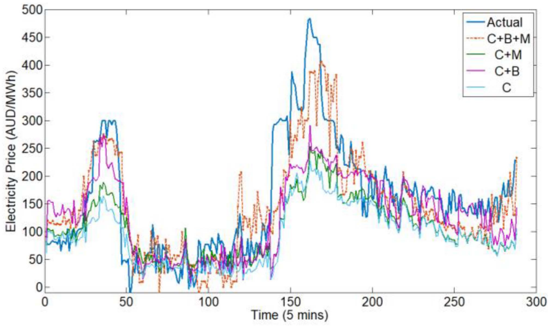


Fig. 8. Day-ahead price predictions with different agent-assisted CTSGAN.

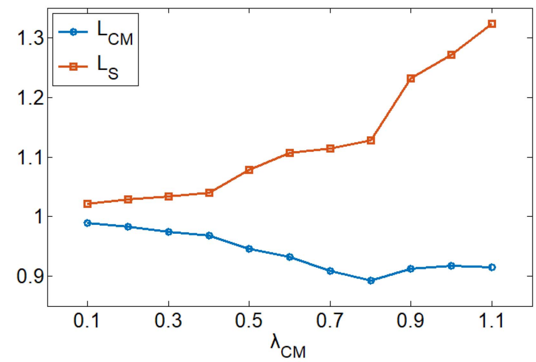
TABLE II
NORMAL PRICE PREDICTION ERRORS OF ABLATION EXPERIMENTS

	RMSE (AUD/MWh)	MAE (AUD/MWh)
C + B + M	58.08	45.45
C + M	84.99	63.10
C + B	72.25	50.58
C	92.62	69.34

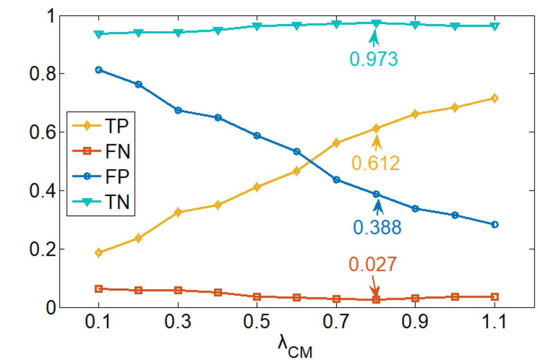
used here is the basic CTSGAN without spike-enhanced component. Fig. 8 presents the comparison between the predicted results and the actual electricity prices for May 21, 2024. The predictions are based on the two-agent assisted CTSGAN (C+B+M), market sentiment agent assisted CTSGAN (C+M), bidding behavior agent assisted CTSGAN (C+B), and basic CTSGAN (C).

It can be observed that the two-agent assisted CTSGAN model closely matches the actual price curve in many periods, especially during significant fluctuations between 150 and 200. The bidding behavior agent-assisted CTSGAN model captures price fluctuations to some extent, reflecting market supply and demand changes. The market sentiment agent-assisted CTSGAN model performs better around 40 than around 160, indicating that the high prices near 40 are more driven by market sentiment, while the high prices near 160 are driven by a combination of market sentiment and bidding behavior. Due to the absence of market sentiment and bidding behavior information, the basic model significantly underperforms the other models in terms of price prediction accuracy.

Table II shows the prediction errors of the ablation experiments in terms of RMSE and MAE for different CTSGAN models. The two-agent assisted CTSGAN model achieves the lowest RMSE of 58.08 and MAE of 45.45, indicating its superior performance in accurately predicting electricity prices. The market sentiment agent-assisted CTSGAN model has a higher RMSE of 84.99 and MAE of 63.10. The bidding behavior agent-assisted CTSGAN model achieves an RMSE of 72.25 and an MAE of 50.58, indicating moderate accuracy. The basic CTSGAN model shows the highest RMSE of 92.62 and MAE of 69.34, demonstrating the least accuracy among the models due to the absence of additional agent information. These results highlight the importance of incorporating both market sentiment and bidding behavior agents to enhance prediction accuracy.



(a)



(b)

Fig. 9. Impact of different λ_{CM} on prediction error: (a) supervised loss and confusion matrix loss, (b) components of the confusion matrix.

D. Verification of Spike Prediction With Spike-Enhanced CTSGAN and Two LLM Agents

When constructing (10), we introduce a weight parameter, λ_{CM}, which balances the supervised loss L_S and the confusion matrix loss L_{CM}. Setting this parameter properly is crucial for the predictive performance of the proposed method. If λ_{CM} is set too high, the focus of the model will shift entirely to identifying the presence of spikes rather than accurately forecasting prices. Conversely, if λ_{CM} is too low, the model will ignore the spike component, causing the proposed spike-enhanced CTSGAN to revert to a conventional CTSGAN, with significantly reduced accuracy in spike prediction. In this section, we conducted experiments regarding λ_{CM} to compare the L_S and L_{CM}, and determine the optimal λ_{CM} for electricity spike price (>600AUD/MWh) predictions.

Fig. 9(a) shows the impact of different choices of λ_{CM} on the supervised loss L_S and the confusion matrix loss L_{CM}. It can be observed that the prediction of spikes becomes increasingly accurate until $\lambda_{CM} = 0.8$, where L_{CM} reaches a minimum value of 0.89. Further increasing λ_{CM} leads to an increase in L_{CM}, indicating that the error starts to rise. The reason can be seen in (b): with the increase of λ_{CM}, TP rapidly increases, meaning the model's prediction of True Positives increases, while FP rapidly decreases. However, FN rises after reaching its minimum value with 0.027 when $\lambda_{CM} = 0.8$, indicating that the prediction model starts to incorrectly predict some non-spikes as spikes. This also explains the increase in L_S after

TABLE III
SPIKE PRICE PREDICTION ERRORS OF ABLATION EXPERIMENTS

	TP	FN
E+ B + M	61.25%	2.65%
E + B	60.21%	2.77%
E + M	54.73%	2.74%
E	52.17%	3.43%
C	18.64%	6.33%

TABLE IV
COMPARISON OF ELECTRICITY PRICE PREDICTION WITH DIFFERENT PREDICTION MODELS

	Proposed	CTSGAN	LSTM	GRU	AEMO
RMSE	47.28	79.43	94.06	108.45	147.92
MAE	38.54	72.92	88.36	91.87	137.11
Spike (TP)	58.95%	23.16%	18.94%	22.11%	N/A

$\lambda_{CM} = 0.8$ in Fig. 9(a). Therefore, we select 0.8 as the optimal weight parameter λ_{CM}.

Table III presents the ablation study results for spike prediction. E represents spike-enhanced CTSGAN, while other abbreviations are the same as in Table II. We compared TP, which represents the proportion of spikes correctly predicted by the model out of all spikes, and FN, which indicates the proportion of non-spikes incorrectly predicted as spikes by the model out of all non-spikes. It can be seen that the spike-enhanced CTSGAN, combined with the bidding behavior agent and market sentiment agent, achieves a TP prediction rate of 61.25% and reduces the FN rate to 2.65%. Additionally, relying solely on the spike-enhanced module can increase the TP rate to 52.17%, compared to the basic CTSGAN's TP rate of only 18.64% for 5-minute spike predictions. Among the two LLM agents, it is evident that the bidding behavior agent improves the TP rate more effectively than the market sentiment agent, although there is no significant difference in reducing the FN rate.

E. Comparative Simulation of Electricity Price Prediction With Different Models

Finally, we compare our proposed LLM agent-assisted spike-enhanced CTSGAN with the basic CTSGAN, LSTM, and GRU. Additionally, we introduce the official price predictions from AEMO. AEMO's prediction model is a half-hour prediction model that rolls every 5 minutes. To compare it with our 5-minute day-ahead prediction model, we select AEMO's prediction data for the next 24 hours at 4:00 am and use interpolation to convert the half-hour predictions into 5-minute predictions. The comparison is based on RMSE, MAE, and spike TP rate. We use all price data from May 2024 as the prediction set and data from October 2021 to April 2024 as the training set. In this prediction set, there are a total of 95 spikes exceeding 600 AUD/MWh. Table IV presents the comparison results of the experiments.

In terms of RMSE, our proposed model performs the best with a value of 47.28, indicating the smallest prediction error. CTSGAN has an RMSE of 79.43, LSTM 94.06, GRU 108.45, and AEMO's official forecast has the largest error at 147.92. Regarding MAE, our model also excels with a value of 38.54, again showing the lowest prediction error. For the TP rate of spikes, our proposed model significantly outperforms the others with a TP rate of 58.95%. CTSGAN has a TP rate of 23.16%, LSTM 18.94%, and GRU 22.11%. Since AEMO only provides half-hour prediction data, it is not possible to obtain 5-minute resolution spike predictions. Therefore, the TP rate of AEMO's predictions is not applicable. Overall, Table IV shows that our proposed prediction model performs the best in terms of RMSE and MAE, indicating the lowest prediction errors. Additionally, our model has a significantly higher TP rate for predicting high price spikes compared to the other models.

VI. CONCLUSION

This paper proposed a spike-enhanced CTSGAN for day-ahead electricity price prediction with a resolution of 5 minutes. The spike-enhanced component effectively increases the TP rate of spike predictions while reducing the FN rate. We further integrated fine-tuned LLaMa3 8B as the bidding behavior agent and ChatGPT-4o as the market sentiment agent. The former agent is used to predict the bidding behavior of electricity trading participants for the next day, refined with a 5-minute resolution. The latter agent analyzes the impact of market sentiment on hourly electricity prices for the next day. Case study shows our model significantly outperformed the predictions provided by AEMO. The RMSE is reduced from 147.92 AUD/MWh to 47.28AUD/MWh, and the MAE is reduced from 137.11AUD/MWh to 38.54AUD/MWh.

As bidding strategies adapt to real-time market fluctuations, network congestion will have economic implications by influencing both dispatch decisions and price dynamics. In future work, we plan to consider the effects of congestion-related factors in our model to better understand how market constraints impact pricing and participant strategies, aiming to refine predictive accuracy under varying system conditions. Integrating such factors will allow for a more holistic approach to capturing the economic drivers behind bidding and price volatility.

Currently, large-scale commercial ESSs aimed at arbitrage are being gradually deployed in Australia. This will inevitably have a greater impact on wholesale electricity prices and intensify market competition. In the future, the high-frequency electricity clearing process will make the wholesale electricity market more similar to other high-frequency trading markets. Long-term factors such as coal prices will merely form the market fundamentals, while short-term factors will play a more significant role in price fluctuations. Therefore, analyzing other participants' bidding behavior and market sentiment will become increasingly important in the future.

APPENDIX

A. Example of Input for Fine-tuning Bidding Behavior Agent Input:

"Tomorrow is October 8, 2021. Based on the historical data for the electricity market participant with DUID of …, please predict this participant's BA for the next day. Assume that tomorrow's PB will be the same as today's.

On October 1, 2021:

PB were $\underbrace{[-1000, -100, 1, 10, 50, 100, 150, 300, 600, 1200]}_{10}$.

For PB1, BA1 values were

$\underbrace{[10,\ 10,\ 10,\ 20,\ 30,\ 40,\ \ldots,\ 10]}_{288}$.

For PB2, BA2 values were […].

…

For PB10, BA10 values were […].

Electricity prices recorded: $\underbrace{[123,\ 124,\ \ldots,\ 68]}_{288}$.

Demand values recorded: $\underbrace{[1200,\ 1300,\ \ldots,\ 1000]}_{288}$.

On October 5, 2021:
PB were […].
For PB1, BA1 values were […].
For PB2, BA2 values were […].
…
For PB10, BA10 values were […].
Electricity prices recorded: […].
Demand values recorded: […].

On October 6, 2021:
…

On October 7, 2021:
…"

Output:

"For PB1, BA1 values are $\underbrace{[15, 15, 10, 25, 35, 45, \ldots, 15]}_{288}$.

For PB2, BA2 values are […].

…

For PB10, BA10 values are […]."

B. Word Cloud for Market Sentiment

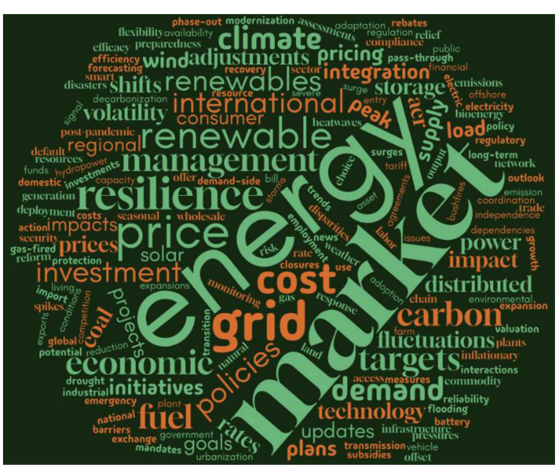


Fig. 10. Word cloud for market sentiment news.

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